TABLE I
DIRECT-C QUERIES AND CROWN CERTIFICATE-CHECKING COSTS FOR THE Q7.8 GTSRB MODEL.

(a) Direct-C queries: validation image 3652

Configuration	Solver	Elapsed (s)	Peak RSS (GiB)	Outcome
Whole network; concrete input	Z3	600	2.78	Timeout
L0: 24 outputs; concrete input	Bitwuzla	80	4.15	Verified
L0: 24 outputs; 12 free bytes	Bitwuzla	186	4.38	Verified
L0: 24 outputs; 18 free bytes	Bitwuzla	270	4.57	Out of memory
L0: 48 outputs; concrete input	Bitwuzla	122	4.99	Out of memory

(b) Completed CROWN checks: full-network certificates, $\varepsilon = 1$ byte

Image (split)	Jobs	Shard wall (h)	Summed elapsed (h)	ESBMC checks	Peak RSS (GiB)	Outcome
06267 (test; primary)	4	3.26	11.82	48,871	4.10	Certified
04083 (test; disclosed rerun)	12/4	2.17	20.48	50,851	8.43	Certified
3652 (validation; development)	4	3.85	13.92	58,695	3.97	Certified

Notes. Direct-C queries use compact int16 storage; CROWN certificates bind to the int64-storage reference C. L0 denotes layer 0. Each direct-C row is one query, with a 600 s timeout and `5g` memory limit; layer queries check interval containment, not classification robustness. CROWN checks use Z3 with 60 s and `4g` per harness. For image 04083, layers 0–3 used 12 jobs and layer 4 used 4. Shard wall is the sum of layer-shard wall times; summed elapsed is the sum of individual check durations, not CPU time. CROWN RSS is the peak aggregate across concurrent checking processes. Finder/export time, preceding box proofs, and unsuccessful earlier attempts are excluded. Certification assumes the binding and composition described in the methodology.